

Lab 04 Inheritance

Submission guidelines

1. Write your answers on the paper and submit.
2. Copy your Java code and submit on Moodle

1. [25 points] **Employee** and **ProductionWorker** Classes:

- a) Design a class named **Employee**. The class should keep the following information in fields:
- Employee name
 - Employee number in the format XXX-L, where each X is a digit within the range 0–9, and the L is a letter within the range A–M.
 - Hire date

Write one or more constructors and the appropriate accessor and mutator methods for the class.

- b) Next, write a class named **ProductionWorker** that inherits from the **Employee** class. The **ProductionWorker** class should have fields to hold the following information:
- Shift (an integer)
 - Hourly pay rate (a double)

The workday is divided into two shifts: day and night. The shift field will be an integer value representing the shift that the employee works. The day shift is shift 1, and the night shift is shift 2.

Write one constructor with all arguments and one without any , and the appropriate accessor and mutator methods for the class.

- c) Demonstrate the classes by writing a program that uses a **ProductionWorker** object.

Sample output:

Here's the first production worker.

Name: John Smith

Employee Number: 123-A

Hire Date: 11-15-2005

Shift: Day

Hourly Pay Rate: \$16.50

Here's the second production worker.

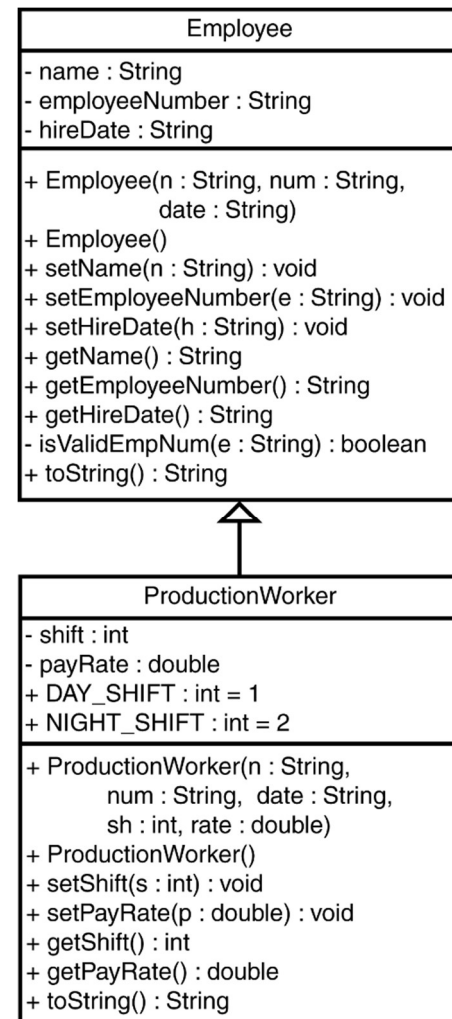
Name: Joan Jones

Employee Number: 222-L

Hire Date: 12-12-2005

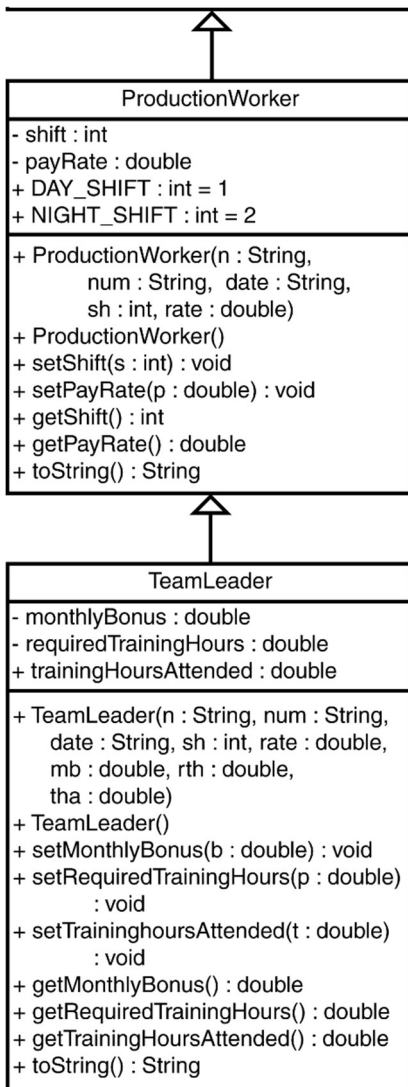
Shift: Night

Hourly Pay Rate: \$18.50



2. [25 points] TeamLeader Class:

In a particular factory, a team leader is an hourly paid production worker who leads a small team. In addition to hourly pay, team leaders earn a fixed monthly bonus. Team leaders are required to attend a minimum number of hours of training per year.



- a) Design a **TeamLeader** class that inherits from the **ProductionWorker** class you designed in the previous exercise. The **TeamLeader** class should have fields for
- the monthly bonus amount,
 - the required number of training hours, and
 - the number of training hours that the team leader has attended.

Write one constructor with all arguments and one without any , and the appropriate accessor and mutator methods for the class.

- b) Demonstrate the class by writing a program that uses a **TeamLeader** object.

Sample output:

```
Here's the first team leader.
Name: John Smith
Employee Number: 123-A
Hire Date: 11-15-2005
Shift: Day
Hourly Pay Rate: $16.50
Monthly Bonus: $500.00
Required Training Hours: 5.0
Training Hours Attended: 2.5
```

```
Here's the second production worker.
Name: Joan Jones
Employee Number: 222-L
Hire Date: 12-12-2005
Shift: Night
Hourly Pay Rate: $18.50
Monthly Bonus: $600.00
Required Training Hours: 7.0
Training Hours Attended: 3.5
```